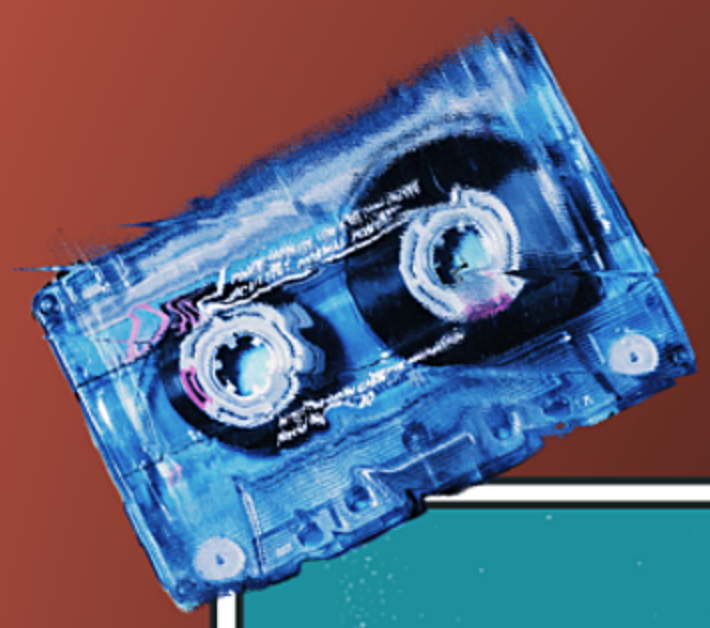
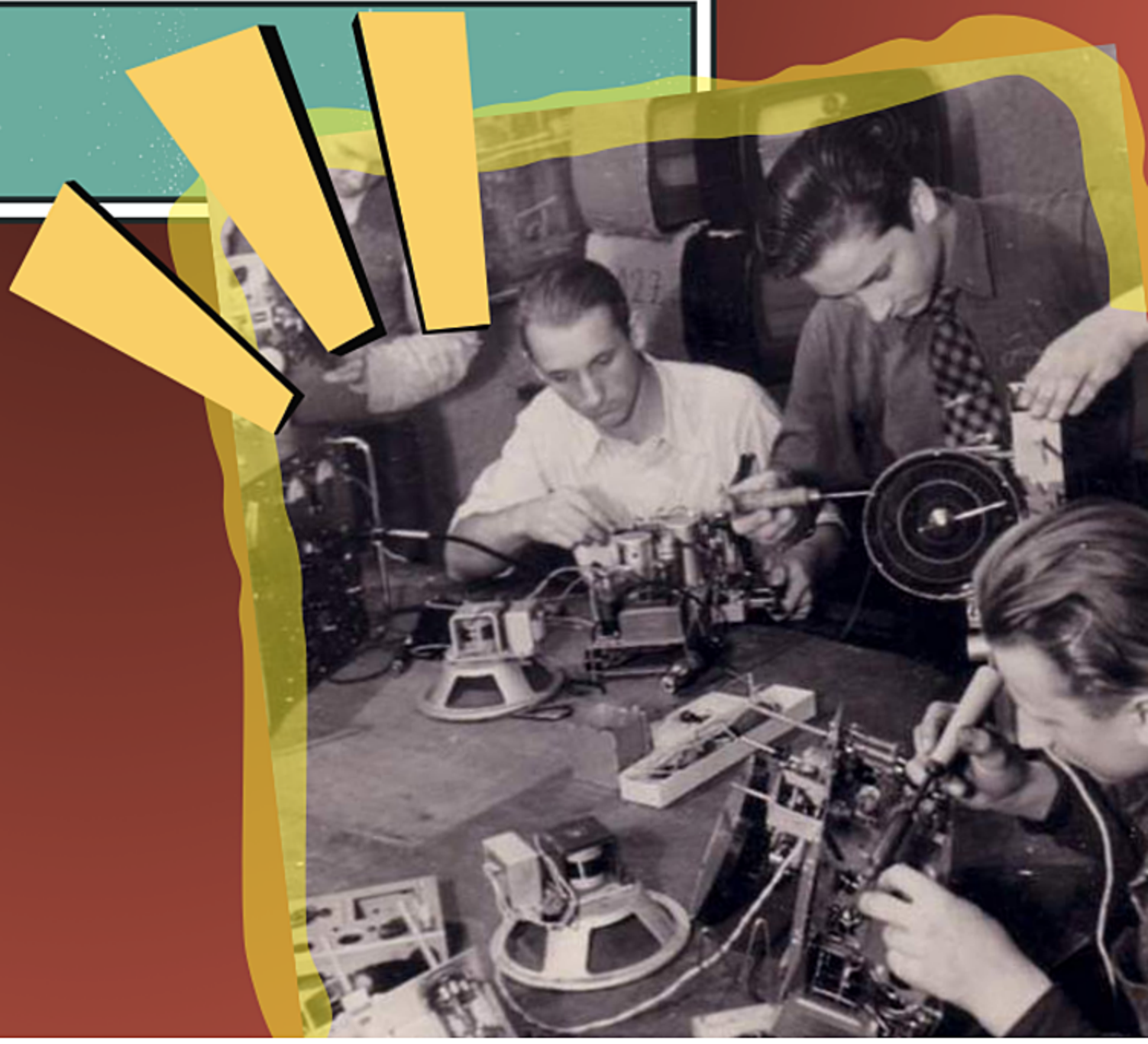

Radio
Electronics

WHAT IS IT?



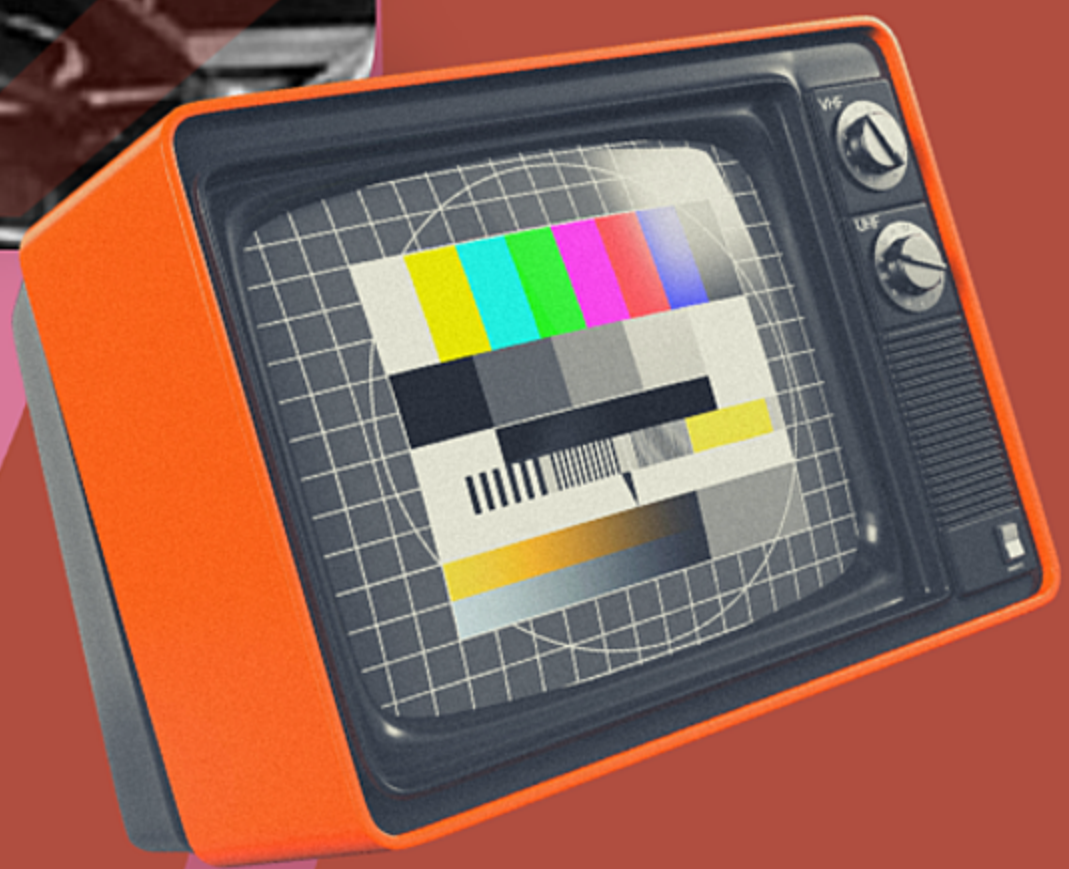
IT IS A FIELD OF SCIENCE AND TECHNOLOGY THAT DEALS WITH THE STUDY AND PRACTICAL APPLICATION OF METHODS FOR TRANSMITTING, RECEIVING, AND CONVERTING INFORMATION USING ELECTROMAGNETIC OSCILLATIONS AND WAVES.




SIMPLY PUT, IT ENCOMPASSES EVERYTHING RELATED TO THE USE OF RADIO WAVES, LIGHT, ELECTRICITY, AND OTHER ELECTROMAGNETIC PHENOMENA FOR DATA EXCHANGE.
But, what did it give to us?



Long-distance transmission of information and electronic data processing In other words, humans were able to transmit and convert energy, which gave a tremendous boost to the development of science



RADAR AND REMOTE SENSING



RADIO ELECTRONICS HAS ENABLED US TO DETERMINE CERTAIN CHARACTERISTICS OF AN OBJECT WITHOUT PHYSICALLY INTERACTING WITH IT. FOR EXAMPLE: A RADAR TRANSMITS A RADIO SIGNAL, RECEIVES ITS REFLECTION, AND USES THE INFORMATION CONVEYED TO DETERMINE SPEED, LOCATION, AND OTHER PARAMETERS. RADAR SYSTEMS USED IN AVIATION, METEOROLOGY, AND SPACE EXPLORATION ARE BASED PRECISELY ON THIS PRINCIPLE.

AND FINALLY.....

Transistors!!

A TRANSISTOR IS A COMPONENT THAT ALLOWS A SMALL INPUT SIGNAL TO CONTROL A SIGNIFICANT CURRENT IN THE OUTPUT CIRCUIT; ITS INVENTION PROVIDED A HUGE BOOST TO TECHNOLOGICAL DEVELOPMENT, SOLVED MANY PROBLEMS–SUCH AS REPLACING VACUUM TUBES–AND SERVED AS A STARTING POINT FOR NEW DEVICES AND FOR EXPANDING THE CAPABILITIES OF CIRCUITS.



How radio electronics relates to transistors?

Before the advent of transistors, vacuum tubes were the foundation of radio electronics; they had serious drawbacks: they consumed a lot of energy, generated a great deal of heat, were fragile and had a short lifespan, and the devices themselves were bulky.

However, in 1947 at Bell Labs, John Bardeen, Walter Brattain, and William Shockley created the first working transistor. It could perform the same functions as a vacuum tube (amplifying and switching signals), but was more compact, more energy-efficient, more reliable, and did not require warm-up.

To sum up.....

Radio electronics is a fascinating branch of science that has forever changed the world of electronics. Components developed back in the days of the first radio receivers are still in use today, and some have even been improved and are now part of our everyday lives.

Thank you!!

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